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Formative evaluation of the online version of the strengthening families program in Brazil

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Abstract
Online family drug prevention programs have been shown to affect the behavior of children and adolescents. This study presents the formative evaluation results of the online implementation of Familias Fortes, a Brazilian-adapted version of the Strengthening Families Program (SFP 10–14). A total of 31 families participated in the evaluation, and data were collected from 74 observations, 64 fidelity forms, 20 focus groups, and 16 semi-structured interviews in groups located in five cities with a total of 113 participants. The evaluation focused on three outcomes: fidelity, appropriateness, and feasibility. Low fidelity was observed, with videos and audio content not consistently sent as planned, although most core elements were maintained. The program's appropriateness showed good accessibility and utilization of video content by participants, but confidentiality was lacking in adolescent meetings. Feasibility analysis revealed challenges related to poor internet access, malfunctioning of facilitator cameras, issues with time management and internal organization of institutions, and insufficient facilitator training. Specific problematic activities were identified, and recommendations for improvement were made. This included enhancing internet connectivity, developing a new manual specific to the online format, improving time management for facilitators, providing more interactive facilitator training, and considering a hybrid format. Adaptations for the aforementioned difficulties are needed before an online version of Familias Fortes could be disseminated in Brazil as a pilot model.

Keywords Formative evaluation, Family-based prevention programs, Virtual programs, Implementation research, Alcohol, Drugs

Background

Adolescent drug use is a significant public health challenge, contributing to the global burden of disease [1, 2]. The 2019 PeNSE (National School Health Survey) reported that 22.6% of Brazilian students aged 13 to 17 had tried cigarettes and 63.3% had tried alcoholic beverages. One in five Brazilian adolescents has consumed alcohol before the age of 13 [3]. Findings from a recent longitudinal study underscore the complexity of substance use trends among adolescents in Brazil during the COVID-19 pandemic, highlighting a nuanced landscape where previous mental health conditions and behavioral risk factors correlate with increased alcohol consumption and behavioral disorders [4].

To address this issue, prevention programs targeting adolescents have shown promise, with family-based interventions demonstrating the most significant evidence of effectiveness [5]. In Brazil, the Programa Familias Fortes (PFF), adapted from the Strengthening Families Program (SPF 10–14), aims to prevent the early onset of

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alcohol consumption among adolescents [6]. The SPF is recognized as one of the most effective interventions for preventing alcohol use among adolescents [7] and is recommended by the UNODC for low- and middle-income countries [8]. Its logical model emphasizes socioemotional and parenting skills to reduce the risk of behavioral problems and drug use among youth [9]. A randomized controlled trial that evaluated the effect of the PFF in Brazil six months after the program application reveals that it effectively enhances nonviolent parenting practices and reduces episodes of parental drunkenness, although it shows no immediate impact on adolescent drug use, suggesting a need for longitudinal evaluations to assess sustained behavioral changes [10].

The COVID-19 pandemic made in-person programs infeasible [11], leading to increased interest in virtual prevention programs [12]. Meta-analyses of studies carried out before covid pandemic showed that virtual family education programs positively impacted parent-child interactions and children's behavior [13]. In response to the cessation of in-person family interventions, the Ministry of Women, Family, and Human Rights (MMFDH) proposed an online version of the PFF [6].

Method

The study aims to evaluate the implementation of a governmental family-based drug use prevention program for adolescents in its online version through a formative evaluation design. This qualitative approach seeks to improve the program during its development or early implementation phase by identifying strengths, weaknesses, and areas for adaptation. The ultimate goal is to provide actionable feedback to government stakeholders to guide necessary adjustments and enhance the program's chance of being effective and relevant [14].

Study design

We performed a formative evaluation of the online version of the PFF, developed by the Brazilian MMFDH, using the framework proposed by Stetler et al. [14], which includes three subtypes of formative evaluation: (1) implementation-focused, (2) progress-focused, and (3) interpretative. The implementation-focused evaluation identified discrepancies between planned and actual implementation through facilitator forms. The progress-focused evaluation assessed family satisfaction and engagement post-implementation, using observations, WhatsApp groups, and focus groups. The interpretative evaluation combined previous data with interviews to analyze the program's sustainability and identify needed adaptations, with interviews conducted until data saturation was reached. The intervention was implemented in five facilities across five Brazilian states (São Paulo, Rio de Janeiro, Mato Grosso do Sul, Acre, and Rio Grande do Norte), including one Social Assistance Reference Center, two Civil Society Organizations, and two Educational Federal Institutes. A total of 113 participants from 53 families (56 guardians and 57 adolescents) in 11 groups (five families and two facilitators each) who underwent the program were observed. The study collected qualitative with 139 observations of WhatsApp Groups and virtual meetings, 64 fidelity forms answered by the facilitators, 20 focus groups with participants, and 16 semi-structured interviews conducted with facilitators and managers of the program.

The approach to presenting the outcomes was discussed during the data analysis phase, and three implementation outcomes were selected as defined by Proctor et al.: fidelity, which refers to the program being delivered as intended and the quality of program delivery; appropriateness, which pertains to the perceived fit and compatibility of the program with the participants; and feasibility, which relates to the actual fit of the program and if the program is suitable and compatible with day-to-day organization of the institution [15]. Specific activities of the program that required alterations were also analyzed. This study followed the guidelines of the Consolidated Criteria for Reporting Qualitative Research (COREQ) [16].

Participants selection

The institutions were selected by the MMFDH. The criteria for selecting these institutions were that they must have at least three professionals duly trained by the MMFDH to deliver the program and the necessary material to conduct online meetings (manuals, supplementary materials, computers with cameras, microphones, and internet connections). The institutions oversaw selecting participating families with one parent or guardian and one child/adolescent (10–14 years old) from the same family willing to participate in the program and available to participate in WhatsApp video meetings. The evaluation performed was observational. The research team did not engage in participant selection but accepted them as determined by the collaborating institutions. No compensation was provided for participation in the study.

Intervention

Face-to-face version

The face-to-face version of the Brazilian PFF (and SFP 10–14) consists of seven 2-hour meetings held weekly. Participants, including guardians and children, attend separate workshops during the first hour and come together as a family in the second hour. These sessions are structured by a videotape that shows videos depicting real-life family situations and provides intervals for discussions, games and dynamics on the topics depicted on video [17]. The program follows a logic model and includes themes and activities outlined in Table 1. A logic model of the program is available on the Blueprint website [18]. According to Rocha et al. [19] the core elements of the program are the (1) themes and sequence of the meetings; (2) use of the video support provided in the program; (3) content activities; (4) duration and format of the activities; (5) role of the facilitators in the conduction of the meetings and in the social support provided; (6) facilitation of a support group formed between the participants; (7) logistic support; and (8) family profiles for participating. The program underwent cultural adaptation in 2014–2015 [19] and has become a public policy in Brazil. The program is currently under the jurisdiction of the Ministry of Justice, with its evaluation being carried out in partnership with the Ministry of Human Rights and Citizenship.

Online version

To adapt to COVID-19 restrictions, an online version of the program was developed in 2020 by the federal government. It maintains material and activities of the face-to-face version but is conducted through WhatsApp. WhatsApp was chosen given I wide usage in Brazil, with numerous telephone companies providing free videocall usage without additional charges. Participants receive handbooks and support materials, such as paper and pencils, to use during the online sessions. WhatsApp groups are created, allowing facilitators to share program information. The videos that structured the face-to-face sessions were sent to the participants in the group, to be watched before the sessions, following a predetermined schedule. Weekly WhatsApp video calls were used for online workshops, with facilitators guiding activities orally and displaying materials through their cameras, sometimes referring to the provided handbook. Participants were requested to keep their cameras open. The online sessions followed the same themes, activities, and durations as the face-to-face versions, except for the videos, which were sent via WhatsApp in the online version. Each group consisted of five families and two facilitators due to WhatsApp's video call limitations. The program spanned seven weeks, with three blocks of meetings: an adolescent call (50 min), a guardian call (30 min), and a family call (30 min), totaling 110 min of facilitator time. The difference in duration between the face-to-face and online sessions was due to the videos being sent to the

Table 1 Meeting themes and mediating variables

Meeting	Participants	Theme	Mediating variables
1	Guardians	Love and limits (Amor e limites)	Clear communication of rules and substance use expectations; empathy with stressors
	Adolescents	Having goals and dreams (Ter metas e sonhos)	Making good decisions/setting goals for the future
	Families	Supporting goals and dreams (Apoiar metas e sonhos)	Family bonding/Affective quality
2	Guardians	The rules of our home (As regras da nossa casa)	Active listening and effective communication
	Adolescents	Admiring mothers, fathers, and caregivers (Admirar as mães, pais e responsáveis)	Empathy and appreciation of parents
	Families	Admiring family members (Admirar os membros da família)	Family bonding/Affective quality
3	Guardians	Encouraging good attitudes (Incentivar boas atitudes)	Reward good behavior
	Adolescents	Dealing with stress (Lidar com o estresse)	Healthy coping and stress management
	Families	Family moments (Momentos em família)	Value time together/family fun
4	Guardians	Using consequences (Usar consequências)	Identify and deliver appropriate consequences calmly
	Adolescents	Following the rules (Seguir as regras)	Understanding the value of rules and responsibilities
	Families	Understanding family values (Compreender os valores da família)	Identify family strengths/values
5	Guardians	Building bridges (Construir pontes)	Monitor youth
	Adolescents	Handling peer pressure (Lidar com a pressão dos amigos)	Peer pressure resistance; understand the harmful impact of problem behavior and substance use
	Families	Strengthening family communication (Fortalecer a comunicação familiar)	Effective communication
6	Guardians	Protecting against substance abuse (Proteger contra o abuso de substâncias)	Clear communication of rules and substance use expectations
	Adolescents	Peer pressure and good friends (Pressão dos amigos e bons amigos)	Know quality of good friends
	Families	Family and peer pressure (Família e pressão dos amigos)	Joint problem solving
7	Guardians	Helping and being helped (Ajudar e ser ajudado)	Monitor youth
	Adolescents	Achieving our goals (Atingir nossas metas)	Making good decisions/setting goals for the future
	Families	Putting it all together (Juntando tudo)	Value time together/family fun

participants prior to the session in the online version, rather than being watched during the session as in the face-to-face version.

Training of the facilitators

Facilitators for both the face-to-face and online versions undergo training provided by the MMFDH. The 25-hour distance-learning course is divided into three modules: an introductory theoretical module, a practical module covering program delivery procedures, and a final module focusing on booster sessions [20]. Trainees complete online evaluations, and upon successful completion, become certified PFF facilitators. The MMFDH is responsible for facilitator training and program implementation.

Procedures and data collection

Data collected from August to December 2021 included data collected during program application (implementation observations and fidelity forms), near program completion (focus groups with participants) and after the program application (interviews with facilitators). Four researchers with prevention program evaluation experience collected and analyzed the data [XX (MD), XX (PhD), XX (PhD), and XX (professor)]. Research materials were developed based on program manuals and literature addressing common barriers and facilitators associated with online prevention programs [21, 22]. Weekly research meetings were held during data collection. Four researchers participated in the data analysis.

Observations of the implementation

During program delivery, researchers observed workshops in two ways: WhatsApp chat group interactions and weekly virtual meetings. Observation of the implementation collected implementation indicators (percentage of the activities that were correctly realized, order of the activities, core components ect). During their observations, the researchers had access to a detailed activity description form. This form was used to identify which activities were completed by the implementer and which were not. Researchers then evaluated each activity on a 5-point scale, ranging from "fully agree" to "fully disagree," to indicate whether the activity was carried out according to the description. Fidelity was considered low when more than half of the observations fell into the three lowest categories. In chat groups, researchers observed facilitators' video and audio content, participant homework, and group participation. All WhatsApp chat groups were observed weekly. In virtual meetings, researchers observed videocall quality, technical issues, and workshop activities. Observations were recorded in specific forms. Each group received at least 4 virtual meeting observations.

74 observations were made in WhatsApp chat groups. 65 observations were made in WhatsApp virtual meetings. Data from these observations measured mainly intervention fidelity but were also triangulated with other outcomes in the results.

Fidelity forms

Facilitators completed weekly fidelity forms with questions about meeting activities, participant numbers, video call quality, and specific workshop activities. Data collected in the fidelity forms was objective. Facilitators were requested to respond to inquiries regarding which scheduled activities were conducted during the session and to what extent these activities were delivered. Fidelity forms also measured participation, providing the number of participants present in each session. Participation was considered sufficient if at least 2 families (from the 5 enrolled) participated in each session. A total of 64 questionnaires were obtained. Data from the forms were analyzed along with other sources to assess implementation fidelity.

Semi-structured interviews with facilitators

After implementation, 16 semi-structured interviews were conducted with managers ($n = 6$) and facilitators ($n = 10$). After the program's completion, facilitators and managers were interviewed by the research team. These interviews, which adhered to a semi-structured guide, lasted approximately one hour. Participants were not compensated for their involvement. The questions posed during the interviews centered on the participants' perceptions of the barriers and facilitators to program implementation, particularly in the online format. Topics that were not initially included in the original guide but emerged during the first interviews were subsequently incorporated into the questions for further exploration. Data evaluated intervention feasibility, including program impact on operations, comfort with activities and virtual format, and suitability for institutions. Interviews were recorded and transcribed for analysis. Data collected in the semi-structured interviews was qualitative.

Focus groups with participants

Near program completion, participants joined focus groups. 20 focus groups were conducted, involving 18 adolescents and 33 adults (10 focus groups each). Participants were informed at the outset that focus groups would be conducted, and consent was obtained through consent forms. Participation in the focus groups was not mandatory for receiving the program. The focus groups, which lasted approximately one hour, were conducted with participants from the same group of program application and were guided by two researchers from the research team. The questions concentrated on participants' perceptions of program activities and their relevance to daily issues. No compensation was provided for participation in the study. Topics that were not initially included in the original guide but emerged during the first focus groups were subsequently incorporated into the questions for further exploration. Data collected from the focus groups were used to evaluate the appropriateness of the intervention. Focus groups were recorded and transcribed for analysis. Data collected in the focus groups was qualitative.

Data analysis

We employed triangulation in the analysis, utilizing observations, fidelity forms, interviews, and focus groups. Fidelity forms and observations were analyzed descriptively, while content analysis was conducted on interviews and focus groups using Bardin's theoretical framework [23]. Axial coding was applied, linking categories based on the interview guide topics and their properties and dimensions [23, 24]. Fidelity was measured by descriptive analysis of objective data provided by observation of the implementation and fidelity forms. Feasibility was evaluated by qualitative analysis of the semi-structured interviews and appropriateness was evaluated by qualitative analysis of the focus-groups.

Results

Six program managers were interviewed for data collection. Five of them were woman, and one was a man, with ages ranging from 38 to 52 years. Ten facilitators were also interviewed (three men and seven women), with ages ranging from 23 to 49 years. Interviews were conducted with one manager and two facilitators from each institution. Eighteen adolescents aged 10–14 years and 33 adults participated in the focus groups.

A total of 53 families were enrolled in 11 groups. One group had two families, one group had three families, eight groups had five families, and one group had 10 families (sessions were held in Google meetings in this group to include all the interested participants). Fifty-six guardians were registered (50 mothers, two grandmothers, and four fathers), and for three families, two guardians were present (mother and father). Fifty-seven adolescents aged 10–14 years participated in the study. In the workshops, there was an average participation of three families in the adolescent, parent, and family workshops. Participation was thus deemed sufficient. Researchers have noted that other family members (including fathers, grandparents, aunts, and younger and/or older children) often participate in sessions although they are not formally enrolled.

The results of the methods applied based on the implementation variables are presented below. Table 2 summarizes the qualitative results and data sources. Table 3 presents the main alterations suggested to address the difficulties encountered.

Fidelity/core elements

In 75% of the workshops, the facilitators considered that they had sufficient time to carry out all the proposed activities. The facilitators declared that they had carried out an average of 86% of the planned activities in each workshop: 90% in the children's workshop, 87% in the caregivers' workshop, and 79% in the families' workshops. However, the researchers' observations revealed low fidelity in the application of certain activities.

In 79% of the groups, all videos were correctly sent following the schedule. In contrast, audio clips were correctly sent in only 36% of the groups. Only 47% of adult participants completed the tasks they were supposed to complete throughout the meeting weeks, with a significant drop over the weeks (from 68% in the first week to 20% in the fifth week). The study also evaluated whether and how the program's core elements were affected by the online version. This research suggests that seven of the core elements of the program were maintained in the online version: themes and sequence of the meetings were respected (except for one activity called "Special Effects," detailed later); participants engaged with the distribution of the videos on their cellphones weekly; duration, format, and content of the activities was respected; the facilitators believe that their link with the families facilitated improvement; the logistics of attending the meetings was respected; and the profile of the families was also maintained. However, one core element was negatively affected, that is, the online version did not facilitate the creation of a support group for participants.

Appropriateness

Confidentiality

The confidentiality of the adolescents' workshops appeared in the evaluation as an obstacle. During the meetings, researchers observed difficulty in maintaining the confidentiality of the workshops as the adults were often in the same room as the adolescents. A suggestion for improvement is to highlight this shortcoming during the facilitators' training, reminding them to watch out for this concern and whenever an adult is present, to kindly ask them to respect this rule and not be in the same room.

Accessibility

In the focus groups, the participants stated that the online version improved accessibility for those who would not have been able to guarantee face-to-face participation, given the difficulties some would have faced in traveling to the institution for meetings. This topic was spontaneously brought up by participants. The accessibility of

Table 2 Synthesis of the results and data source

Topic	Subtopic (data source)	Results and Golden quotes
Fidelity/Core elements	Time for application of activities, proportion of activities performed, homework (fidelity forms)	Activities carried out in full, but with low fidelity in application
	Core elements kept (observations and interviews of researchers)	Most core elements kept. One not kept: the virtual version did not allow the creation of a support group among the participants. *"The proximity with the families augmented significantly during the program application" (Interview G1)* *"Our relationship with the parents was very different after the program" (Interview G1)*
	Sending videos and audios (observations)	Videos sent correctly; audios sent incompletely in general
Appropriateness	Accessibility (focus groups)	*"If it wasn't the online program I would not have been able to participate" (focus groups adults 5)* *"It's impossible to organize myself to go every week to the institution" (focus groups adults 6)*
	Videos on WhatsApp (focus groups)	*"It was good to be able to watch the videos from home" (focus groups adults 3)* *"I would sometimes watch the videos multiple times; I even sent them to some friends" (focus groups adults 3)*
	Confidentiality (researchers' observations)	Workshop confidentiality not respected (presence of other family members in the same room during the workshops)
Feasibility	Internet access (researchers' observations and interviews)	The course of the meetings much more impacted by the connection problems of the facilitators than of the participants. *"When we had internet connection problems it was really difficult to continue the meeting" (Interview F3)*
	Presentation of materials through the camera (researchers' observations)	Activities that require the facilitator's camera to be used to show something did not work
	Institutional Organization (interviews)	*"When we are doing the program, all of our other activities in the week are impacted by it" (Interview F5)* *"The ideal would be to have a team only dedicated to the application of the program" (Interview G4)*
	Too short a time for complex activities (researchers' observations)	Workshops with many consecutive short activities presented greater difficulties in execution.
	Facilitators' training (interviews)	*"The training is insufficient for the facilitators that have more difficulties" (Interview G1)* *"We need more experienced facilitators to be able to help the ones that are beginning" (Interview G5)*
Activities	"Special effects" activity (researchers' observation)	Facilitators suggesting that drug use is equivalent to having superpowers.
	"Listing Small Household Tasks" activity (focus groups)	*"He has to know that living in a house he must share obligations like everyone else and that shouldn't be rewarded" (focus groups adults 2)*
	Activities: "Supporting Children's Dreams and Goals" and "How Your Parents Treated You" (researchers' observations and focus groups)	Activities that lead many adults to relive difficult memories of their own childhoods. *"We have our childhood traumas, I had many (…) we need to make a difference (…) when this was dealt with in the program, I saw myself in those stories and thought 'I need to change'"* (*focus groups* adults 2)

sending of the videos via WhatsApp in the online version was perceived in a positive way by the participants.

Video materials

In the virtual version, videos are sent gradually through the WhatsApp chat group to be watched throughout the week, and there is interaction with facilitators through message exchange. The adult participants positively perceived the distribution of content and interaction with facilitators over time. However, on one specific week (between meetings 5 and 6), the participants said that they had received too many videos and did not have time to watch all of them. We suggest that some of the videos could be removed or, in meeting six, facilitators might give participants time to watch the videos.

Feasibility

The feasibility of the program was affected by various factors, including low-quality internet connections, difficulties in presenting materials through cameras, internal organization problems within institutions, insufficient time for complex activities, and inadequate facilitator training provided by the MMFDH platform.

Table 3 Recommendations

Category	Difficulty Encountered	Suggested Solution
Fidelity	Failure to form a support group among participants (Core element)	Organize informal in-person meetups for participants, such as coffee and snack breaks
Appropriateness	Too many videos sent between meetings 5 and 6, hindering effective utilization of materials	Remove some of the videos or schedule dedicated time to watch them at the beginning of the next meeting
	Lack of confidentiality in the workshops	Highlight the importance of maintaining confidentiality during the facilitators' training
Feasibility	Internet connectivity issues, particularly for facilitators, significantly impacting the progress of the meetings	Make a reliable internet connection an absolute priority for participating institutions
	Activities requiring the facilitator's camera to present materials are not functioning	Short-term: Verbally convey the phrases or describe the images to participants instead of showing them through the camera Medium-term: Include the materials (phrases, images) in the participants' workbook
	Difficulty in organizing institutionally for program implementation, demanding excessive time from facilitators	Adapt the workload of professionals working as facilitators in the program to ensure they are not overwhelmed and do not negatively impact overall institutional functioning
	Insufficient time for complex activities	Reorganize the workshops to have fewer activities, but allocate more time to each activity
	Inadequate training to adequately equip facilitators	Include an opportunity for facilitators to interact with experienced facilitators responsible for the program during their training
Activities	"Special Effects" activity associates drugs with superpowers	Modify the example of superpowers in this activity or remove it completely
	"Small Household Tasks" activity associates domestic chores with punishment	Remove domestic chores from the points system
	"Supporting Children's Dreams and Goals" and "How Your Parents Treated You" activities trigger difficult childhood memories	Include a warning in the facilitator's manual about the potential for intense emotional reactions from participants during these activities, along with suggestions for facilitators on how to manage these situations

Internet access

Facilitators faced difficulties connecting to the internet and experienced poor video call quality. These issues disrupted the sessions, particularly in groups with few participants, where interruptions due to internet failure significantly affected group dynamics. To address this, it is recommended that institutions prioritize ensuring a high-quality internet connection for facilitators when implementing the online version of the program.

Presenting materials through the camera

The activities where the facilitator was supposed to present material led to recurring execution problems. The facilitators' cell phone cameras did not provide clear images, causing technical difficulties and impacting the activity's development and fidelity. In groups that avoided using the cell phone camera and instead read aloud the content that should be shown, fewer problems were encountered. In the short term, it is suggested to simplify these activities by conducting them through spoken phrases without attempting to show content through a cell phone camera. In the long term, modifying the adult and adolescent activity books to include necessary images and phrases for these activities is recommended.

Internal organization within institutions

Scheduling conflicts for facilitators arose, as their workload extended beyond the two-hour weekly sessions. They were also responsible for sending audios and videos, answering participants' questions, and assigning homework. This was a major challenge reported by managers and facilitators. The program at one institution was postponed for several weeks due to scheduling conflicts and challenges faced by the municipality. During the interview, the facilitators reported that this delay was due to difficulties in reconciling their schedules with the program's demands. To address this issue, it is recommended to provide institutions with an estimate of the time needed to implement the program and prioritize time management for facilitators during the application weeks. Additionally, establishing a dedicated team solely responsible for program implementation would be ideal for long-term continuity.

Insufficient time for complex activities

Many short activities were often completed incompletely or incorrectly when performed in sequence. In workshops with fewer planned activities, they were more thoroughly accomplished. It is suggested that the program's manual be revised by the MMFDH to maintain the main activities while reorganizing and combining several smaller activities into longer, cohesive ones whenever possible. This approach would provide clarity and enhance the effectiveness of the application.

Facilitator training

Managers and facilitators felt that the training offered by the MMFDH platform was insufficient for correct program implementation. Lack of interaction with a trainer or tutor to address specific questions and the absence of reports from experienced facilitators were highlighted as drawbacks. To improve training, it is recommended to review the facilitator training program and incorporate mentoring by experienced facilitators. If direct mentoring is not feasible, including videos that depict practical examples of errors in program implementation and demonstrate expected outcomes would be beneficial.

Addressing these feasibility challenges will contribute to the successful implementation and effectiveness of the program. By prioritizing internet access, revising activity organization, improving facilitator training, and ensuring effective internal organization within institutions, the program can overcome obstacles and deliver better outcomes.

Specific activities that presented problems

Four program activities require attention:

"Special effects" activity

In the fifth meeting of the adolescents' workshop, this activity aimed to teach them not to blindly trust television. However, facilitators in multiple groups changed the topic and compared it to using drugs with special powers. To address this, either change the example or completely revise the activity. Program managers suggested a hybrid version with face-to-face moments to facilitate support group formation.

"Listing domestic chores" activity

In the fourth meeting of the caregivers' workshop, the activity introduced a "point system" where adolescents gain or lose points based on their behavior and activities. Domestic chores were included as possible consequences of negative acts. However, some parents expressed discomfort with associating domestic activities with punishment. To address this, remove domestic activities from the point system or clearly explain their role and importance in the family routine.

"Supporting children's dreams and goals" and "How your parents treated you" activities

These activities occur in the first and fifth meetings of the caregivers' workshop. Participants reflect on their upbringing, their relationship with their parents, and what they would change to avoid repeating negative experiences with their children. Some participants expressed difficulty due to their past experiences. While considering the impact on mental health, the activities had a positive effect overall. Retain the activities but add a note to the facilitators' script about potential effects and provide more time. Ideally, offer support options like a Psychosocial Care Center (CAPS).

Addressing these problematic activities is crucial for improving the program's effectiveness and participant experience. By making appropriate revisions and providing support where needed, the program can better meet participants' needs and expectations.

Discussion

This study presents the findings of a formative evaluation conducted on the virtual version of the PFF, focusing on fidelity, appropriateness, and feasibility of implementation. The results found major challenges in the application of the program in online format. Nevertheless, several benefits of the online format were observed, such as improved accessibility, access to video content, and enhanced proximity between facilitators and families.

Improved accessibility is a noteworthy aspect of the online version, as previous studies have highlighted that the absence of the need to travel to an institution is one of the main strengths of online interventions [25]. Telehealth care, in particular, eliminates access difficulties typically associated with limited opportunities or transportation systems [26, 27]. In the context of Brazil, where the size of the country and the distance between people and services can pose challenges, enhancing accessibility to social and health programs is crucial. Adapting the program to an online version not only benefits families during the current pandemic but also enables long-term dissemination, providing opportunities for those who face limitations in terms of working hours, lifestyle, or transportation.

The results related to program feasibility highlight several implementation challenges that need to be addressed for wider dissemination of the intervention. While the possibility of remote participation improves accessibility, it is essential to have good quality internet connections to facilitate engagement in video calls [28]. Institutions applying the program experienced more difficulties with connectivity compared to individual participants. Ensuring facilitators have reliable internet access and minimizing their connection issues during sessions are crucial. Implementation plans must consider providing participating institutions with reliable connections from both technical and financial perspectives, as observed in other countries where similar programs were adapted to online formats [13, 14].

Regarding fidelity of implementation, the evaluation focused on the impact of the online version on the formation of support groups, a central element of the program

[9]. To address this challenge, managers suggested adopting a hybrid program format that combines online sessions with informal in-person meetings organized by the institutions during the program's duration. This hybrid learning format, proven to be promising in the international literature [29], could help overcome difficulties in forming support groups online. Additionally, using culturally adapted and up-to-date video materials is crucial, as feasibility studies for the face-to-face version of the program have already highlighted this issue [30].

Low fidelity in implementing the program's activities was also observed, and fidelity is considered a key element for achieving expected preventive effects in interventions [31]. The debate between ensuring reliable application of the program and making local adaptations to fit different contexts is ongoing [32]. Recent research suggests finding a balance between modifications to the original model to suit the new context while maintaining "functional loyalty" to the program's core elements [33]. Enhancing fidelity requires revising facilitator training to emphasize the importance of maintaining core elements while preparing facilitators for necessary adaptations in practice [34]. Similar challenges with fidelity have been observed in face-to-face prevention programs [35], necessitating adequate training for facilitators to achieve higher fidelity in application while adapting to different scenarios [30].

While the health crisis drove the online adaptation of the program, the concept of online interventions, especially in the health field, has been a subject of interest for researchers for years [36]. Telehealth, defined as remote healthcare without physical contact via telephone or internet-based communication methods [37, 38], has the potential to continue beyond the crisis. In the post-COVID-19 era, institutions implementing the program could maintain both face-to-face and online services, offering a hybrid version that combines the benefits of both approaches.

It is important to acknowledge the limitations of this study. The research sample was not designed to be representative, and future program applications may vary in terms of participant groups and composition. The feedback provided to the program's responsible institutions regarding evaluation difficulties and adaptation proposals may lead to different outcomes in future evaluations.

Conclusion

In conclusion, the online version of the PFF shows significant potential, offering key advantages such as improved accessibility and the integration of video materials. However, challenges related to internet connectivity, facilitator training, and fidelity to the program's core elements must be addressed before broader implementation. Strategies such as providing technical support, enhancing training to balance fidelity with contextual adaptation, and exploring hybrid formats can strengthen the program's feasibility and impact. By overcoming these barriers, the online PFF could become a valuable tool for expanding access to social and health services, supporting its adoption as a public policy initiative.

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Authors' contributions

ZS secured fund acquisition and study coordination. TP and ZS conceived the analysis and collected the data. All authors contributed to data analysis. TP, RG, and JY wrote the manuscript. RG, JY reviewed and revised the manuscript. ZS critically revised the manuscript.

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Data availability

The datasets used and analyzed during the current study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate

Ethics approval (0662/2021 CAAE: 47881821.0.0000.5505) was obtained from the UNIFESP Research Ethics Committee. Participants were informed about the study and provided written consent or assent through a form, in accordance with the Declaration of Helsinki.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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